

SECURESIGHT

Digital Media Forensic Analysis Report

ISO/IEC 27037:2012 · ISO/IEC 27041:2015 · ISO/IEC 27042:2015

Report Reference	SR-A1B2C3D4
Evidence ID	EV-2024-00042
Filename	suspect_video_interview.mp4
Report Date	13 August 2026 — 00:37 UTC
Classification	CONFIDENTIAL
Examiner	Dr. J. Smith, CFCE — Digital Forensics Laboratory
Examination Method	Automated Multi-Pipeline AI & Forensic Analysis
Software Version	SecureSight v2.0
Report Standard	ISO/IEC 27037:2012 / 27041:2015 / 27042:2015

OVERALL ASSESSMENT

LIKELY FAKE

73.4_{/100}

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1. Executive Summary

This report presents the findings of an automated digital media forensic examination conducted on the file 'suspect_video_interview.mp4' (Evidence ID: **EV-2024-00042**). The examination was performed using SecureSight's 15-pipeline analysis framework, which employs deep learning models, image forensic algorithms, biometric analysis, and physical consistency checks to determine the authenticity of digital media.

OVERALL AUTHENTICITY SCORE

73.4 / 100

Verdict: **LIKELY FAKE**

The submitted media exhibits multiple characteristics consistent with AI-generated or digitally manipulated content. Strong indicators were detected across multiple independent detection pipelines, suggesting the media has been artificially produced or significantly altered.

Score Range	Verdict	Interpretation
0 – 15	AUTHENTIC	High confidence genuine media
15 – 35	LIKELY AUTHENTIC	Probably genuine, minor anomalies
35 – 60	SUSPICIOUS	Warrants further investigation
60 – 85	LIKELY FAKE	Strong manipulation indicators
85 – 100	CONFIRMED FAKE	Near-certain AI/deepfake content

2. Scope & Methodology

This examination was conducted in accordance with the following international standards:

Standard	Title	Compliance
ISO/IEC 27037:2012	Digital Evidence — Identification, Collection, Acquisition & Preservation	✓ Full
ISO/IEC 27041:2015	Assuring Suitability and Adequacy of Investigation Methods	✓ Full
ISO/IEC 27042:2015	Analysis and Interpretation of Digital Evidence	✓ Full
SWGDE v3.0	Best Practices for Digital & Multimedia Evidence	✓ Partial
NIST SP 800-86	Guide to Integrating Forensic Techniques into Incident Response	✓ Partial

Examination Methodology

The analysis employs a multi-layered detection framework consisting of 15 independent detection pipelines organised into 4 tiers. Pipelines that cannot run (e.g. face-only models on images with no detected faces, or video-only pipelines on static images) are **excluded from scoring** — they do not vote 'authentic'. The final score is derived from a calibrated weighted ensemble of only the pipelines that actively ran.

Tier	Pipelines	Focus Area
Tier 1 — Deep Learning	EfficientNet-B4, AI Ensemble (ViT+CLIP+SigLIP), XceptionNet, AlexNet, InceptionV3	AI-driven image/speech-based classification
Tier 2 — Image Forensics	ELA, Copy-Move, JPEG Ghost, EXIF Metadata	Pixel-level manipulation detection
Tier 3 — Physical	Eye Reflection, Shadow/Lighting, Noise Pattern	Physical world law violations
Tier 4 — Advanced	Frequency Domain, Biometric Face Mesh, Temporal	Spectral, geometric & temporal anomalies

3. Evidence Description & Integrity Verification

The following digital evidence was submitted for examination. Cryptographic hash values were computed at the time of intake to ensure evidence integrity throughout the examination process, in compliance with **ISO/IEC 27037:2012 §7.1.3**.

Property	Value
Filename	suspect_video_interview.mp4
Evidence ID	EV-2024-00042
Analysis Ref.	SR-A1B2C3D4
Intake Timestamp	2026-08-13 00:37:30 UTC
SHA-256 Hash	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
SHA-512 Hash	cf83e1357eefb8bdf1542850d66d8007d620e4050b5715dc83f4a921d36ce9ce47d0d13c5d85f2b0ff8318d2877eec2f63b931bd47417a81a538327af927da3e
Integrity Status	✓ VERIFIED — Hash values computed and recorded at intake

4. Chain of Custody

The following chain of custody log documents all actions performed on the evidence, in compliance with **ISO/IEC 27037:2012 §7.2**. Each entry includes the action performed, the actor, timestamp, and cryptographic hash.

#	Timestamp (UTC)	Action	Actor	Hash (first 16 chars)
1	2024-11-02 09:15:44	Evidence Intake	analyst@lab.gov	e3b0c44298fc1c14...
2	2024-11-02 09:15:45	Hash Computed	system	e3b0c44298fc1c14...
3	2024-11-02 09:15:46	Analysis Dispatched	system	e3b0c44298fc1c14...
4	2024-11-02 09:16:22	Analysis Complete	system	e3b0c44298fc1c14...
5	2024-11-02 09:16:23	Report Generated	system	e3b0c44298fc1c14...

5. Detection Pipeline Results

The following table presents results from each active detection pipeline. Scores range from 0.000 (strongly authentic) to 1.000 (strongly manipulated). Pipelines are sorted by tier and score. Pipelines that did not run for this media type are omitted.

Pipeline	Tier	Score	Confidence	Time (ms)	Flag
Efficientnet	1	0.821	0.94	312	✗ Significant
Ai Ensemble	1	0.756	0.91	1204	✗ Significant
Xception	1	0.689	0.87	198	✗ Significant
Exif	2	0.650	0.85	12	✗ Significant
Ela	2	0.512	0.80	45	■ Minor flag
Jpeg Ghost	2	0.340	0.75	67	■ Minor flag
Copy Move	2	0.120	0.70	89	✓ No anomaly
Eye Reflection	3	0.780	0.88	23	✗ Significant
Noise Analysis	3	0.410	0.78	56	■ Minor flag
Shadow Lighting	3	0.290	0.72	34	✓ No anomaly
Biometric	4	0.610	0.82	145	✗ Significant
Frequency	4	0.540	0.76	78	■ Minor flag

Key Findings (Score > 0.50):

Pipeline	Score	Detail
Efficientnet	0.821	TTA votes: 5
Eye Reflection	0.780	—
Ai Ensemble	0.756	Fusion: weighted_vote
Xception	0.689	—
Exif	0.650	Software: Adobe Photoshop 2024
Biometric	0.610	—
Frequency	0.540	—
Ela	0.512	—

6. Visual Evidence & Heatmap Analysis

■ No heatmap visualisations were generated for this analysis. Heatmaps are produced when the EfficientNet-B4 or XceptionNet model runs with GradCAM enabled.

7. EXIF Metadata Analysis

The following EXIF metadata was extracted from the submitted evidence file. Anomalies such as stripped metadata, editing software tags, and timestamp inconsistencies are flagged as potential indicators of manipulation.

Metadata Tag	Value
Image Make	Apple
Image Model	iPhone 14 Pro
Image Software	Adobe Photoshop 24.0 (Windows)
EXIF DateTimeOriginal	2024:03:15 14:22:31
EXIF DateTimeDigitized	2024:11:02 09:15:44
GPS GPSLatitude	[37, 46, 29.88]
GPS GPSLongitude	[122, 24, 47.08]
EXIF ExifImageWidth	4032
EXIF ExifImageLength	3024
Image XResolution	72
Image YResolution	72
EXIF ColorSpace	1
EXIF FocalLength	6.86
EXIF ISOSpeedRatings	64
EXIF ShutterSpeedValue	8.644

8. Findings & Interpretation

In accordance with **ISO/IEC 27042:2015 §8**, the following interpretation is provided based on the aggregated results of all active detection pipelines:

The submitted media exhibits multiple characteristics consistent with AI-generated or digitally manipulated content. Strong indicators were detected across multiple independent detection pipelines, suggesting the media has been artificially produced or significantly altered.

AI Ensemble Analysis (35% weight):

The 3-model AI ensemble (haywoodloan ViT, umm-maybe CLIP, Ateeqq SigLIP+DINOv2) produced an aggregated score of **0.756** using the **weighted_vote** fusion strategy. Voting: 3 AI vote(s), 0 real vote(s), 0 uncertain.

EfficientNet-B4 Analysis (25% weight):

The custom-trained EfficientNet-B4 model produced a score of **0.821** using Test-Time Augmentation (5 variants). This model is the highest single-weight detector in the ensemble.

XceptionNet Analysis (10% weight):

XceptionNet produced a score of 0.689.

9. Conclusions & Limitations

Based on comprehensive automated analysis using 15 independent detection pipelines, the submitted file 'suspect_video_interview.mp4' (Evidence ID: EV-2024-00042) received an overall score of 73.4/100 with a verdict of **LIKELY FAKE**.

Limitations of This Report:

- L1 This analysis is performed by automated systems and should be reviewed by a qualified digital forensic examiner before use in legal proceedings.
- L2 Detection accuracy may be affected by heavy image compression, very low resolution, or novel AI generation methods not present in the training data.
- L3 An AUTHENTIC verdict does not guarantee the media has not been manipulated; it indicates no detectable manipulation was found using current methods.
- L4 A CONFIRMED FAKE verdict is a strong indicator but not conclusive proof; certain legitimate image processing may trigger false positives.
- L5 The AI ensemble models are pre-trained on publicly available datasets and may have biases for specific content types or generation methods.
- L6 Video analysis accuracy depends on sampling rate; subtle per-frame manipulations in undersampled segments may be missed.

10. Certification & Disclaimer

CERTIFICATION OF EXAMINATION

I hereby certify that this report accurately represents the findings of the digital forensic examination conducted on the submitted evidence. The examination was performed using scientifically validated methods in accordance with ISO/IEC 27037:2012, ISO/IEC 27041:2015, and ISO/IEC 27042:2015.

Examiner:	Dr. J. Smith, CFCE — Digital Forensics Laboratory
Date:	13 August 2026
Time:	00:37:30 UTC
Report Ref.:	SR-A1B2C3D4
Evidence ID:	EV-2024-00042
Software:	SecureSight v2.0 — Forensic AI Detection Platform

Signature:

Date:

DISCLAIMER: This report is generated by SecureSight, an automated digital forensic analysis platform. While the system employs state-of-the-art AI models and forensic techniques, the results should be interpreted by qualified forensic examiners. This report does not constitute legal advice. The findings are probabilistic assessments and should not be considered as definitive proof of authenticity or manipulation. Any use of this report in legal proceedings should be accompanied by expert testimony from a certified digital forensic examiner. SecureSight and its operators assume no liability for decisions made based solely on the contents of this report.

Appendix A — Glossary of Terms

Term	Definition
AUC	Area Under the ROC Curve — measures model discrimination (0.5 = random, 1.0 = perfect)
CLIP	Contrastive Language–Image Pre-training — multimodal AI model by OpenAI
Deepfake	AI-generated or AI-manipulated media designed to mimic authentic content
DINOv2	Self-supervised vision transformer by Meta — used in SigLIP ensemble model
ELA	Error Level Analysis — detects manipulation by comparing JPEG compression artefacts
EfficientNet	CNN architecture trained on face crops to classify as real or AI-generated
EXIF	Exchangeable Image File Format — metadata embedded by cameras and editing software
GAN	Generative Adversarial Network — AI architecture that generates synthetic images
GradCAM	Gradient-weighted Class Activation Mapping — neural network explainability method
SHA-256/512	Secure Hash Algorithm — cryptographic hash functions for data integrity verification
SigLIP	Sigmoid Loss for Language-Image Pre-Training — Google/Meta vision-language model
TTA	Test-Time Augmentation — multiple inference passes with varied inputs for accuracy
ViT	Vision Transformer — attention-based architecture for image classification
XceptionNet	Depthwise separable CNN architecture specialised for face manipulation detection

Appendix B — Pipeline Methodology Details

Pipeline	Architecture / Technique	Parameters / Training
EfficientNet-B4	CNN (ImageNet → fine-tuned), TTA ×5	42,930 images, AUC 99.77%, JPEG augmentation
AI Ensemble	ViT + CLIP + SigLIP+DINOv2 (3 models)	Pre-trained on HuggingFace (200K+ downloads)
XceptionNet	Depthwise separable CNN, TTA ×5	Face crop specialisation, custom head
Audio	Mel-spectrogram + CNN	Detects synthesised voice / lipsync artefacts
LipSync	Mouth region temporal + audio correlation	Cross-modal consistency verification
ELA	Dual-quality JPEG recompression (Q95+Q75)	Cross-quality consistency comparison
Copy-Move	ORB keypoints + FLANN matching + RANSAC	3000 keypoints, geometric verification
JPEG Ghost	Compression sweep (Q50–100, step 5)	Block-level variance at optimal quality
EXIF Metadata	Multi-source extraction + anomaly checks	Software tags, thumbnails, timestamps
Eye Reflection	Corneal specular highlight NCC matching	Area ratio, centroid, cross-correlation
Shadow/Lighting	Sobel gradient direction per quadrant	Angular consistency + face asymmetry
Noise Pattern	fastNIMeansDenoising + block variance	64×64 blocks, coefficient of variation
Frequency Domain	FFT azimuthal average + 8×8 DCT	High-freq energy ratio, spectral analysis
Biometric	MediaPipe 468-point face mesh	EAR, symmetry, mouth geometry, jitter
Temporal	Frame-to-frame difference + SSIM	Temporal coherence across video frames

— END OF REPORT —